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MASTER OF SCIENCE IN DATA SCIENCE AND BIG DATA ANALYTICS

COURSEWORK

DATA SCIENCE RESEARCH METHODS AND PUBLICATIONS

RESEARCH PROPOSAL

Design and Development of an Intelligent Credit Risk Prediction and Management Tool Using Machine Learning

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Abstract

Credit risk, the possibility that a borrower fails to meet contractual repayment obligations, is among the most consequential risks faced by banks, microfinance institutions, savings and credit cooperatives, and digital lenders. Conventional assessment based on fixed rules, scorecards and manual judgement is transparent but often struggles to capture non-linear patterns in large and changing datasets. This proposal sets out the design and development of an intelligent, data-driven tool that estimates the probability of loan default using machine learning, classifies borrowers into configurable risk bands, and presents the results through an interactive dashboard and automated reports.

The study follows a design-science approach organised around the CRISP-DM process. Logistic regression, decision trees, random forests and gradient boosting will be trained on publicly available credit datasets and compared using ROC-AUC, precision–recall analysis, F1-score, the Kolmogorov–Smirnov statistic and calibration measures, with attention to class imbalance. Explainability (SHAP) and fairness checks are built into the pipeline so that outputs support, rather than replace, human lending decisions. A modular Python architecture with a Streamlit interface will deliver data validation, prediction, risk categorisation, reporting, logging and model versioning. Design choices are aligned with the Ugandan credit-reference-bureau framework and the Data Protection and Privacy Act, 2019. The expected outcome is a tested prototype and an evidence-based evaluation of its predictive quality, usability and responsible-use safeguards.

Keywords: credit risk; probability of default; machine learning; explainable AI; decision support; dashboard; Uganda.

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List of Abbreviations

AUC	Area Under the (ROC) Curve
BoU	Bank of Uganda
CRB	Credit Reference Bureau
CRISP-DM	Cross-Industry Standard Process for Data Mining
DPPA	Data Protection and Privacy Act, 2019 (Uganda)
EAD	Exposure at Default
EL	Expected Loss
KS	Kolmogorov–Smirnov statistic
LGD	Loss Given Default
ML	Machine Learning
NPL	Non-Performing Loan
PD	Probability of Default
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique

1 Introduction

1.1 Background

Lending is the core business of most financial institutions, and the quality of the loan book largely determines their profitability and stability. Credit risk management is therefore a foundational discipline, formalised internationally in the Basel Committee's principles for the management of credit risk (Basel Committee, 2000) and reflected in loan-origination and monitoring guidance such as that of the European Banking Authority (European Banking Authority, 2020). At its centre is the estimation of the *probability of default* (PD): the likelihood that a borrower will fail to repay within a defined horizon. PD, combined with loss given default and exposure at default, yields the expected loss on an exposure,

$$\mathrm{EL} = \mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD}, \tag{1}$$

which supports pricing, provisioning (for example under IFRS 9, IASB, 2014) and portfolio monitoring.

Traditional credit assessment relies on predefined rules, historical repayment records, application scorecards and manual evaluation (Hand and Henley, 1997; Thomas et al., 2002; Siddiqi, 2017). These methods are interpretable and well understood, but they may find it hard to detect complex interactions in large datasets, to adapt as borrower behaviour changes, and to give timely, consistent insight at scale. Large benchmarking studies show that modern classifiers can outperform the logistic-regression baseline on many credit datasets (Baesens et al., 2003; Lessmann et al., 2015), and that machine learning can add value in consumer-credit settings (Khandani et al., 2010).

1.2 The Ugandan context

Uganda's lending environment has several features that make better risk analytics valuable. A statutory credit-reference-bureau system operates under the Financial Institutions (Credit Reference Bureau) Regulations, 2022 (Republic of Uganda, 2022), with the Bank of Uganda supervising bureau activity and publishing the supporting regulations (Bank of Uganda, n.d.). Additional licensed bureaux have entered the market (Daily Monitor, n.d.), and lenders have been encouraged to raise awareness of the bureau system among borrowers (DFCU Group, n.d.). At the same time, press and regulatory commentary has repeatedly discussed non-performing loans, subdued private-sector credit growth and uncertainty in the banking sector (Daily Monitor, 2024; The EastAfrican, 2024; Eagle Online, 2025), and the IMF's 2026 Article IV consultation again addressed financial-sector conditions (IMF, 2026). Cross-country indicators of non-performing loans to gross loans for Uganda are also published by the World Bank (World Bank, 2024).

Recent applied research has begun to examine credit default prediction with Ugandan data using machine-learning models (Muddu et al., 2026), and work on alternative data, such as mobile-phone usage, shows how predictive signals can extend credit assessment to thin-file borrowers (Björkegren and Grissen, 2020). This proposal builds on that direction by focusing on a complete, deployable decision-support *tool* rather than a model alone.

1.3 Proposed solution in brief

The project proposes an intelligent credit risk prediction and management tool that integrates machine learning, algorithmic decision logic, automated analytics and interactive visualisation within a well-architected software system. The tool will convert borrower and loan data into actionable insight so that institutions can assess creditworthiness, identify high-risk borrowers and make consistent, data-driven, and explainable lending decisions.

2 Problem Statement

Financial institutions need to decide quickly and consistently which applicants to lend to, on what terms, and which existing borrowers need early attention. Three problems recur in practice.

1. **Limited predictive power.** Rule-based and manual assessment often fail to capture non-linear relationships and variable interactions, which leads to avoidable defaults or to the rejection of creditworthy applicants (Lessmann et al., 2015).
2. **Inconsistency and delay.** Manual review varies between officers and branches, and portfolio-level warning signs may be noticed late.
3. **Opacity and fairness risk.** Where complex models are adopted, lenders and regulators still need to understand and justify decisions, and models trained on historical data can reproduce bias (Rudin, 2019; Fuster et al., 2022; Kozodoi et al., 2022).

There is therefore a need for a practical system that combines accurate default prediction with transparent risk categorisation, automation of routine analytical steps, and clear visual reporting, while keeping a human decision-maker in control.

3 Aim, Objectives and Research Questions

3.1 Aim

To design and develop an intelligent, data-driven software tool for credit risk prediction and management using machine learning and automated analytics.

3.2 Specific objectives

O1. Identify and preprocess relevant borrower and loan datasets for credit risk analysis.

O2. Develop and evaluate predictive machine-learning models for estimating the probability of loan default.

O3. Implement algorithmic logic for borrower risk classification and automated decision support.

O4. Design interactive visualisations and dashboards to communicate credit-risk patterns, model predictions and portfolio performance.

O5. Develop a modular, secure and scalable software architecture integrating data processing, predictive modelling, automation and reporting.

O6. Evaluate predictive performance, usability, and the tool's potential contribution to data-driven credit management.

3.3 Research questions

RQ1. Which machine-learning algorithm gives the best discrimination and calibration for default prediction on the selected datasets?

RQ2. How should predicted probabilities be converted into low-, medium- and high-risk categories that are stable and useful to lenders?

RQ3. Which borrower and loan characteristics most influence predicted default, and can these influences be explained to non-technical users?

RQ4. To what extent do the dashboard and automated workflow meet usability and decision-support needs?

Table 1. Alignment of objectives, research questions and evaluation evidence.

Objective	Research question	Evidence
O1	—	Data-quality report; documented preprocessing pipeline
O2	RQ1	ROC-AUC, PR-AUC, F1, KS, Brier score on held-out test data
O3	RQ2	Risk-band table; default rate per band; threshold sensitivity analysis
O4	RQ3	SHAP summaries; dashboard screens
O5	—	Architecture documentation; unit and integration test results
O6	RQ1–RQ4	Model comparison; usability questionnaire; limitations review

4 Scope and Significance

4.1 Scope

The study covers tabular, borrower-level data for binary default prediction (default versus non-default) over a fixed outcome window. Public datasets will be used for development and testing. The system will be a research prototype: it will not make autonomous lending decisions, and live integration with production core-banking systems and credit-bureau interfaces is outside the scope of this phase, although the architecture will anticipate it.

4.2 Significance

- **Practical:** faster, more consistent screening and earlier identification of high-risk accounts.

- **Academic:** a reproducible comparison of classical and ensemble methods, with calibration, explainability and fairness examined together.
- **Local relevance:** an open design that smaller Ugandan lenders could adapt, informed by the national CRB and data-protection frameworks.

5 Literature Review

5.1 Credit scoring foundations

Statistical credit scoring began with discriminant analysis and logistic regression, which remain industry standards because they are stable, easy to validate and straightforward to translate into points-based scorecards (Hand and Henley, 1997; Thomas et al., 2002; Siddiqi, 2017). Their limitation is the assumption of linear, additive effects on the log-odds, so complex interactions must be engineered by hand.

5.2 Machine learning for default prediction

Baesens et al. (2003) and, later, Lessmann et al. (2015) benchmarked many classifiers across credit datasets. The latter found that heterogeneous ensembles and several tree-based methods tended to outperform individual classifiers, while noting that practical gains must be weighed against complexity and interpretability. Yeh and Lien (2009) compared data-mining techniques for predicting credit-card default, and Khandani et al. (2010) showed economically meaningful gains from machine-learning models in consumer credit. Random forests (Breiman, 2001) and gradient boosting (Friedman, 2001; Chen and Guestrin, 2016) are now common choices for tabular data. In contrast, Gunnarsson et al. (2021) examine whether deep learning is worth its added complexity for credit scoring, which supports starting with strong tree-based baselines.

5.3 Class imbalance, evaluation and calibration

Defaults are usually the minority class. Resampling methods such as SMOTE (Chawla et al., 2002) and cost-sensitive learning are widely examined for imbalanced credit data (Brown and Mues, 2012). Accuracy is misleading under imbalance; ROC analysis (Fawcett, 2006) should be complemented by precision–recall analysis (Saito and Rehmsmeier, 2015). Because lenders use predicted values as probabilities for pricing and provisioning, calibration matters as well as ranking (Niculescu-Mizil and Caruana, 2005).

5.4 Explainability, fairness and responsible use

Credit decisions are high-stakes. Post-hoc methods such as SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020) attribute a prediction to input features and have been applied in credit risk (Bussmann et al., 2021). Rudin (2019) argues that interpretable models should be preferred where possible for high-stakes decisions, which motivates keeping a transparent logistic baseline in the comparison. Evidence that machine learning can change who gains or loses access to credit (Fuster et al., 2022), together with formal

fairness criteria such as equality of opportunity (Hardt et al., 2016) and profit-aware fairness methods (Kozodoi et al., 2022), motivates fairness checks in this project.

5.5 Software engineering for ML systems

Models are only one component of a production system. Sculley et al. (2015) describe the hidden technical debt of ML systems (data dependencies, configuration, monitoring), reinforcing the need for modular design, versioning and testing. The study's artefact-building approach follows design-science principles (Hevner et al., 2004).

5.6 Summary and gap

Table 2. Summary of selected prior work and its relevance to this proposal.

Source	Contribution	Relevance here
Lessmann et al. (2015)	Large benchmark of classifiers for credit scoring	Justifies comparing several model families
Brown and Mues (2012)	Imbalanced credit data experiments	Guides resampling and metric choice
Niculescu-Mizil and Caruana (2005)	Probability calibration for supervised learners	Calibration step before risk banding
Lundberg and Lee (2017)	Unified feature-attribution (SHAP)	Explanations in dashboard
Kozodoi et al. (2022)	Fairness in credit scoring	Fairness audit design
Muddu et al. (2026)	ML credit default prediction using Ugandan data	Local evidence; motivates tool-level extension
Sculley et al. (2015)	Technical debt in ML systems	Modularity, monitoring, versioning

Most prior studies focus on model accuracy. Fewer deliver an end-to-end, explainable, auditable decision-support tool that couples prediction with risk banding, automation and reporting. This project addresses that gap.

6 Methodology and System Design

6.1 Research design

The project adopts a design-science approach (Hevner et al., 2004) in which an artefact (the tool) is built and evaluated iteratively. The data-science workflow follows CRISP-DM (Chapman et al., 2000): business understanding, data understanding, data preparation, modelling, evaluation and deployment.

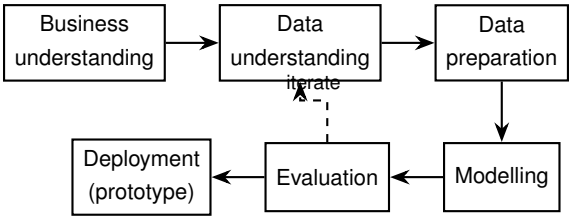


Figure 1. CRISP-DM workflow adopted for the study.

6.2 Data collection and preparation

Publicly available credit datasets will be used for development and testing, since institutional loan data is confidential. Candidates are listed in Table 3. Variables depend on the dataset but typically include borrower income, loan amount, repayment history, credit utilisation, debt-to-income ratio, employment information and previous defaults.

Table 3. Candidate public datasets.

Dataset	Description	Source
Statlog German Credit	1,000 applicants, 20 attributes; small, good for prototyping	Hofmann (1994)
Default of Credit Card Clients	30,000 clients with payment history and bill/payment amounts	Yeh (2016)
Give Me Some Credit	Large retail-credit set with strongly imbalanced default flag	Kaggle (2011)
Home Credit Default Risk	Rich multi-table data for applicants with limited history	Home Credit Group (2018)

Preparation steps are: (i) schema and range validation; (ii) duplicate and outlier handling; (iii) missing-value imputation (median or mode, with missingness indicators where informative); (iv) categorical encoding; (v) feature engineering (ratios such as loan-to-income, payment-delay counts, utilisation); and (vi) feature scaling for scale-sensitive models. To avoid leakage, all fitted transformations are learned on training data only and applied to validation and test data inside a single pipeline.

6.3 Predictive modelling

Candidate algorithms are logistic regression (interpretable baseline), decision trees, random forests (Breiman, 2001) and gradient boosting (Friedman, 2001; Chen and Guestrin, 2016), implemented with scikit-learn (Pedregosa et al., 2011). The logistic model estimates

$$\Pr(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp\big(-(\beta_0 + \beta^\top \mathbf{x})\big)}, \tag{2}$$

where $Y = 1$ denotes default. Data will be split into stratified training, validation and test sets, with k-fold cross-validation for tuning and random search (Bergstra and Bengio, 2012) for hyper-parameters. The test set is used once, for final reporting. Class imbalance will be handled by class weighting and, as a comparison, SMOTE (Chawla et al., 2002) applied only inside training folds. Probabilities will be calibrated (Platt or isotonic) and checked with reliability curves.

6.4 Evaluation metrics

Table 4. Evaluation metrics.

Metric	Purpose
ROC-AUC (Fawcett, 2006)	Overall ranking quality across thresholds
PR-AUC (Saito and Rehmsmeier, 2015)	Performance on the minority (default) class
Precision, recall, F1	Threshold-specific error trade-off; $F_1 = \frac{2PR}{P+R}$
KS statistic	Maximum separation of defaulter and non-defaulter score distributions
Brier score; calibration plot	Accuracy and reliability of probabilities (Niculescu-Mizil and Caruana, 2005)
Confusion matrix	Counts of true/false positives and negatives; cost interpretation

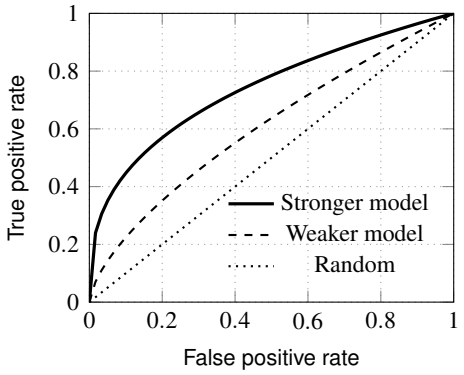


Figure 2. Schematic ROC curves illustrating how models will be compared (illustrative shapes only, not results).

6.5 Risk classification and decision logic

The calibrated PD is mapped to a risk band by configurable thresholds $t_1 < t_2$:

$$\text{Band}(p) = \begin{cases} \text{Low}, & p < t_1 \\ \text{Medium}, & t_1 \le p < t_2 \\ \text{High}, & p \ge t_2. \end{cases} \tag{3}$$

Thresholds will be chosen from validation data by examining default rates per band and the cost trade-off between approving a defaulter and rejecting a good borrower, and will be adjustable by authorised users. Bands *support, not replace*, human decisions: high-risk cases are flagged for review and borderline cases are routed to an officer, as in Figure 3.

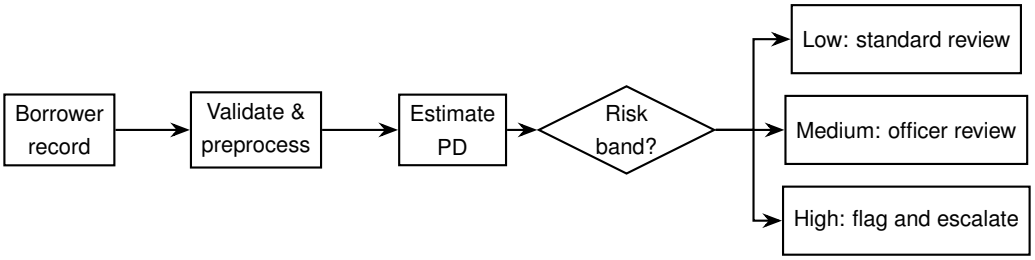


Figure 3. Risk classification and human-in-the-loop decision flow.

6.6 Explainability and fairness checks

Global and local explanations will be generated with SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020). Where protected or proxy attributes are available in a dataset, group-wise error rates and approval rates will be compared using equality-of-opportunity measures (Hardt et al., 2016; Kozodoi et al., 2022). Findings will be reported as limitations and mitigation options rather than as claims of fairness.

6.7 Automation and intelligent analytics

The system will automate data validation, preprocessing, prediction, risk categorisation and report generation; flag high-risk cases; and summarise portfolio trends (risk-band distribution, average PD, movement between bands over time). Human review remains part of consequential lending decisions.

6.8 Visualisation and reporting

An interactive dashboard will present default probabilities, risk-band distributions, important predictive variables, model performance metrics and portfolio trends, with filters by borrower characteristics or loan category. Reports will be exportable for audit.

6.9 Software architecture

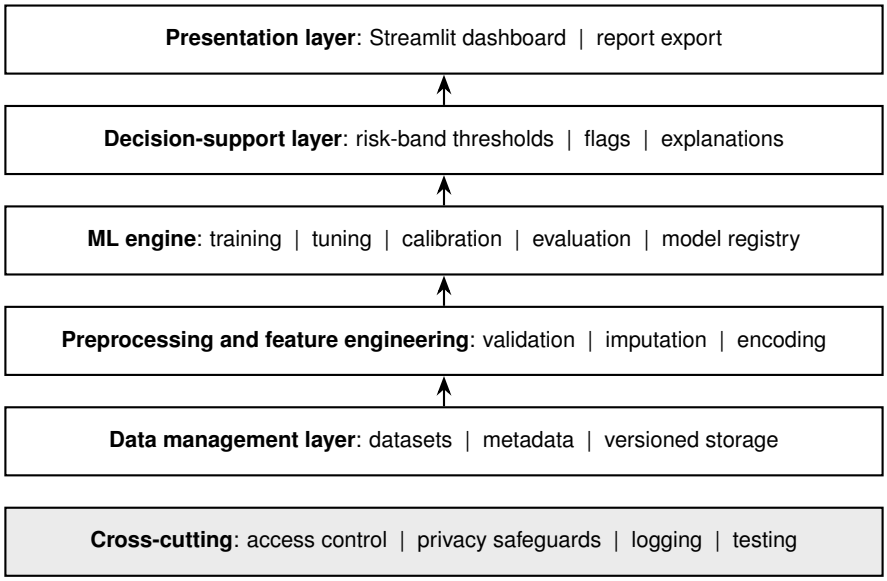


Figure 4. Layered, modular architecture of the proposed tool.

Python supports data processing and modelling; Streamlit or a similar framework provides the interface. Each layer is a separate module with a defined interface so components can be tested, replaced or scaled independently. Access controls, privacy safeguards, audit logging, model versioning and automated tests are included from the outset.

6.10 Evaluation of the tool

Beyond model metrics, the tool will be evaluated on functional correctness (test suite and acceptance checks against stated requirements), performance (response time on typical batch sizes), and usability. Usability will be assessed with a short structured questionnaire and task-based walkthrough with a small group of domain-aware participants, using a standard instrument such as the System Usability Scale.

7 Requirements and Technology Stack

Table 5. Main functional and non-functional requirements.

ID	Requirement
FR1	Upload and validate borrower data against a schema
FR2	Train, compare and store models with versioned metadata
FR3	Return PD and risk band for single and batch records
FR4	Show feature-level explanation for each prediction
FR5	Provide dashboard views and filters; export reports
NFR1	Security: role-based access, protected data at rest and in transit
NFR2	Privacy: data minimisation and purpose limitation in line with Republic of Uganda (2019)
NFR3	Reproducibility: fixed seeds, logged configuration, pinned dependencies
NFR4	Maintainability: modular code with automated tests

Table 6. Proposed technology stack.

Component	Technology
Language	Python
Data and modelling	pandas, NumPy, scikit-learn (Pedregosa et al., 2011), XGBoost (Chen and Guestrin, 2016)
Explainability	SHAP
Interface	Streamlit (or similar)
Testing and quality	pytest, logging, version control (Git)

8 Work Plan

A proposed 20-week schedule is shown in Figure 5; it can be adjusted to the institution's calendar.

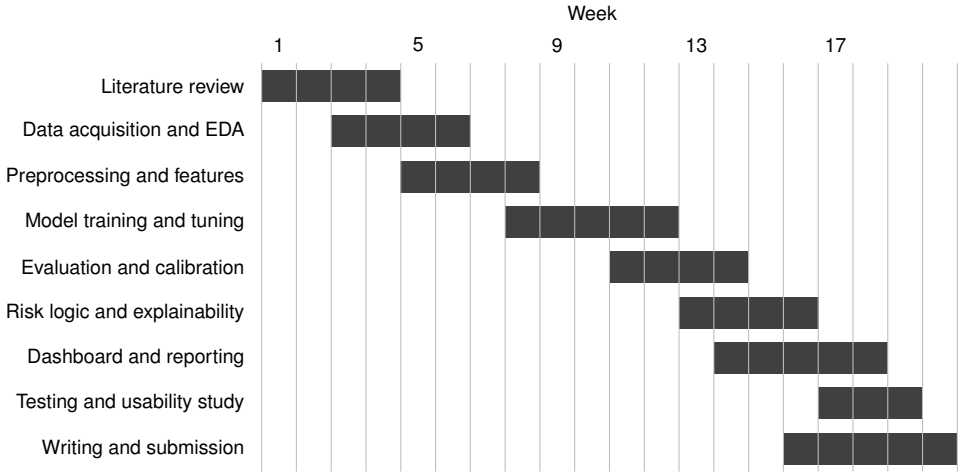


Figure 5. Proposed 20-week work plan.

9 Risks, Ethics and Limitations

Table 7. Key risks and mitigation.

Risk	Mitigation
Public data differs from Ugandan lending portfolios	State limitation clearly; design tool for retraining on local data; compare with local findings (Muddu et al., 2026)
Class imbalance and overfitting	Stratified splitting, cross-validation, appropriate metrics, held-out test set
Biased or opaque predictions	Fairness checks, SHAP explanations, interpretable baseline, human review
Data leakage	Single pipeline fitted on training data only
Privacy breach	Use public or anonymised data; access control; adherence to Republic of Uganda (2019)
Schedule slippage	Prioritised requirements; iterative delivery

Ethics. The study uses publicly available, anonymised datasets and involves no personal data from real borrowers. Any usability participants will give informed consent and remain anonymous. Predictions are decision *support*; final lending decisions stay with accountable humans, consistent with the regulatory emphasis on responsible lending and data protection (Republic of Uganda, 2022, 2019).

Limitations. Historical data may not reflect future conditions, public datasets may not represent local borrowers, and accuracy on one dataset does not guarantee performance on another.

10 Expected Outcomes

1. A functional prototype that analyses borrower data, estimates default probabilities, classifies risk levels, generates visual reports and supports informed lending decisions.

2. A documented, reproducible comparison of candidate models with calibration, explainability and fairness findings.

3. Software architecture and test documentation that can be adapted by other institutions.

4. A final report and, where appropriate, a draft paper for publication.

The project is expected to improve the consistency and efficiency of credit assessment, help identify default risk earlier and strengthen portfolio monitoring, while explicitly considering fairness, interpretability, privacy and the limits of historical data.

11 Conclusion

This project applies data science and artificial intelligence to a practical financial problem by developing a well-architected, intelligent credit risk prediction and management tool. By combining machine learning, algorithmic logic, automated analytics and interactive visualisation, the proposed system provides a practical foundation for data-driven credit assessment and risk monitoring. Its effectiveness will be judged by predictive performance, system functionality and the quality of insight offered to decision-makers.

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A Illustrative Pipeline Code

The following sketch shows the intended structure of the training pipeline (illustrative, not final code).

```
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.calibration import CalibratedClassifierCV
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score, average_precision_score

X_tr, X_te, y_tr, y_te = train_test_split(
   X, y, test_size=0.2, stratify=y, random_state=42)

prep = ColumnTransformer([
   ("num", Pipeline([("imp", SimpleImputer(strategy="median")),
                ("sc", StandardScaler())]), num_cols),
   ("cat", OneHotEncoder(handle_unknown="ignore"), cat_cols)])

model = CalibratedClassifierCV(GradientBoostingClassifier(), method="isotonic",
    cv=5)
clf = Pipeline([("prep", prep), ("model", model)]).fit(X_tr, y_tr)

pd_hat = clf.predict_proba(X_te)[:, 1]
print(roc_auc_score(y_te, pd_hat), average_precision_score(y_te, pd_hat))

def band(p, t1=0.10, t2=0.30): # thresholds set from validation data
   return "Low" if p < t1 else ("Medium" if p < t2 else "High")
```

B Project Quality Checklist

- ☐ Data split performed before any fitting; no leakage
- ☐ Metrics suited to imbalance reported with the test set used once
- ☐ Probabilities calibrated and reliability curve produced
- ☐ Explanations and fairness checks documented
- ☐ Logging, access control and model versioning in place
- ☐ Limitations and ethical considerations stated